Amalia Madden

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Theoretical Physicist and AI Researcher

Experience

Anthropic **June 2026–September 2026**
STEM Fellow (Contract) *San Francisco, CA, USA*

- Design evaluations and RL environments.
- Apply AI to scientific research problems in physics.

Kavli Institute for Theoretical Physics, UC Santa Barbara **August 2024–Present**
Postdoctoral Fellow *Santa Barbara, CA, USA*

- Conduct research at the intersection of particle theory, experimental design, and AI.
- Proposed and led development of FERMIACC, a system of scaffolded LLM agents that searches existing LHC data for evidence of new particle physics models, with deterministic verification of candidates. Awarded $10,000 in API credits from OpenAI for Science.
- Leading a multi-institution effort with Lawrence Berkeley National Laboratory and MIT to design, fund, and fabricate a prototype phonon-based dark-matter detector.

Education

Perimeter Institute for Theoretical Physics **2019–2024**
PhD in Physics *Waterloo, ON, Canada*
Thesis title: New Experimental Observables for the QCD Axion
Supervised by Asimina Arvanitaki

Perimeter Institute for Theoretical Physics **2018–2019**
MSc in Physics *Waterloo, ON, Canada*
Perimeter Scholars International (PSI) Program

Imperial College London **2014–2018**
MSci in Physics, 1st class *London, UK*

Software

Languages & tools: Python (NumPy, Agents SDK); Mathematica; Git/GitHub.

Scientific computing: High-performance computing; development of scientific software and simulations.

Open-source software: Primary contributor to piezoaxionic-effect and ferroaxionic-effect.
Developed software to predict signals from axion dark matter in piezoelectric materials, informing the design of new dark matter and force experiments.

Media and Outreach

OpenAI Academy Blog Feature **March 2026**
How physicists are using AI to chase new physics

Phys.org Coverage **March 2026**
Piezoelectric materials enable new approach to searching for axions

IEEE Central Coast Public Lecture **May 2025**
Dark Matter: Discovering the Invisible Universe

Cafe KITP Public Lecture **April 2025**
Dark Matter: Discovering the Invisible Universe

Perimeter Institute News **July 2024**
Searching for axions

Publications & Preprints

1. N. Craig, **A. Madden**, *"High-quality Axion Strings from Monopole Strings,"* arXiv:2606.16445 [hep-ph] (2026).
2. P. Agrawal, N. Craig, **A. Madden**, I. Valenzuela Lombera, *"The FERMIACC: Agents for Particle Theory,"* arXiv:2603.22538 [hep-ph] (2026).
3. I. M. Bloch, S. Knapen, X. Li, **A. Madden**, G. Marocco, *"Dark Matter Detection Using Phonon Sensing in Amorphous Materials,"* arXiv:2603.22390 [hep-ph] (2026).
4. A. Arvanitaki, J. Engel, A. A. Geraci, A. Hepburn, **A. Madden**, K. Van Tilburg, *"Detecting the QCD Axion via the Ferroaxionic Force with Piezoelectric Materials,"* Phys. Rev. Lett. **136**, 081803 (2026); arXiv:2411.10516.
5. I. M. Bloch, S. Knapen, **A. Madden**, G. Marocco, *"Broadband Phonon Production from Axion Absorption,"* JHEP **03** (2025) 080; arXiv:2411.10542.
6. A. Arvanitaki, **A. Madden**, K. Van Tilburg, *"The Piezoaxionic Effect,"* Phys. Rev. D **109**, 072009 (2024); arXiv:2112.11466.
7. J. Huang, **A. Madden**, D. Racco, M. Reig, *"Maximal Axion Misalignment from a Minimal Model,"* JHEP **10** (2020) 143; arXiv:2006.07379.

Awards and Scholarships

Gordon and Betty Moore Foundation Postdoctoral Fellowship **2026–2027**
UC Santa Barbara *Santa Barbara, CA*

Emmy Noether Emerging Talent Scholarship **2019–2021**
Perimeter Institute *Waterloo, ON, Canada*

Perimeter Scholars International Award **2018–2019**
Perimeter Institute *Waterloo, ON, Canada*

Canada Memorial Foundation Scholarship **2018–2019**
Canada Memorial Foundation *Canada & UK*

President's Undergraduate Scholarship **2014–2018**
Imperial College London *London, UK*

G-Research Prize **2017**
Imperial College London *London, UK*

Selected Talks

SITP Seminar **April 2026**
Stanford University *Stanford, CA, USA*

Dark Universe Science Center Seminar **October 2025**
University of Washington *Seattle, WA, USA*

Particle Seminar **September 2024**
Lawrence Berkeley National Laboratory (LBNL) *Berkeley, CA, USA*

ENS Summer Institute **June 2024**
École Normale Supérieure *Paris, France*

Dalitz Seminar in Fundamental Physics **May 2024**
University of Oxford *Oxford, UK*

High Energy Particle Physics Seminar **November 2023**
University of California San Diego *San Diego, CA, USA*

High Energy Physics Seminar **November 2023**
Caltech *Pasadena, CA, USA*

Referees

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